来伊份 AI 导购助手 HiAgent 架构文档 V1.0

文档信息

项目	内容
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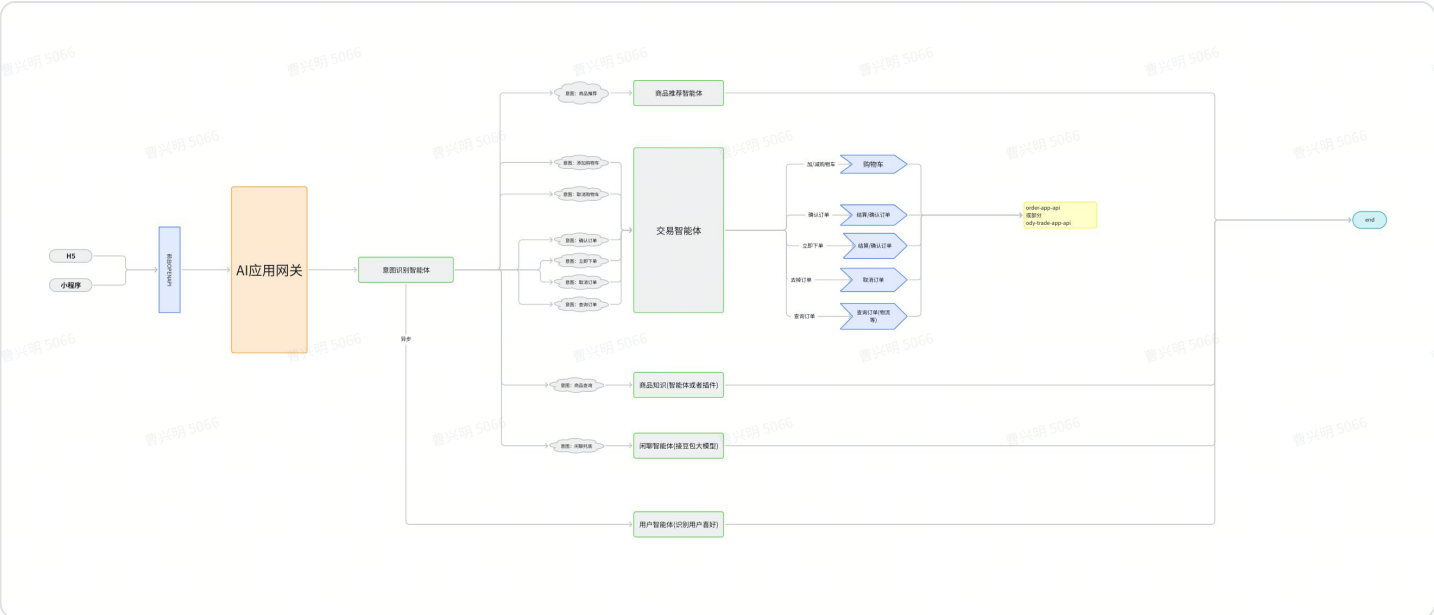
1 架构总览

1.1 系统定位

来伊份 AI 导购助手是基于 HiAgent 平台构建的智能对话导购系统，通过 AI Agent 技术实现用户购物全流程的智能化服务。

一期PRD：Ai 导购助手-第一阶段产品需求文档

1.2 整体架构图



1.3 Agent 与业务中心映射

Agent	职责	交互业务中心
意图Agent	意图识别、槽位补全、统筹协调	调度所有业务中心
交易Agent	加购、删改商品、数量调整	交易中心、商品中心
	下单、支付、价格计算	交易中心、支付中心、营销中心
	订单查询、物流查询	订单中心、履约中心
问答Agent	FAQ问答、商品知识	商品中心、营销中心、用户中心
商推Agent	商品推荐、场景推荐	商品中心、营销中心、用户中心
用户Agent	用户画像、偏好记忆	用户中心
闲聊Agent	豆包生成闲聊回复	豆包大模型

2 整体交互流程

2.0 意图关键词

前置智能体	协作智能体	意图名称(CN)	意图名称(EN)	智能体协作流程
意图识别智能体	商推智能体(1000)	商品推荐	view_items	
	交易智能体(2000) transaction_process	查询购物车	view_shop_cart	获取购物车列表
		添加购物车	add_shop_cart	添加商品到购物
		删除购物车	del_shop_cart	购物车删除删除商品
		清空购物车	clear_shop_cart	清空购物车商品
		确认订单	confirm_order	确认商品信息后，下单操作

		立即下单	place_order_oow	搜索商品，确认订单
		取消订单	cancel_order	当用户指定订单号
		查询订单	query_order	
	知识问答智能体(3000)	知识问答	knowledge_quiz	
	闲聊智能体(4000)	闲聊回复	casual_reply	
用户智能体				

2.1 交互链路

代码块

```
前端 → 网关 → HiAgent(意图Agent) → 业务Agent → 业务中心
                                                ↓
前端 ← 网关 ← HiAgent(意图Agent) ← 业务Agent ← 业务中心

```

2.2 Agent 协作流程

代码块

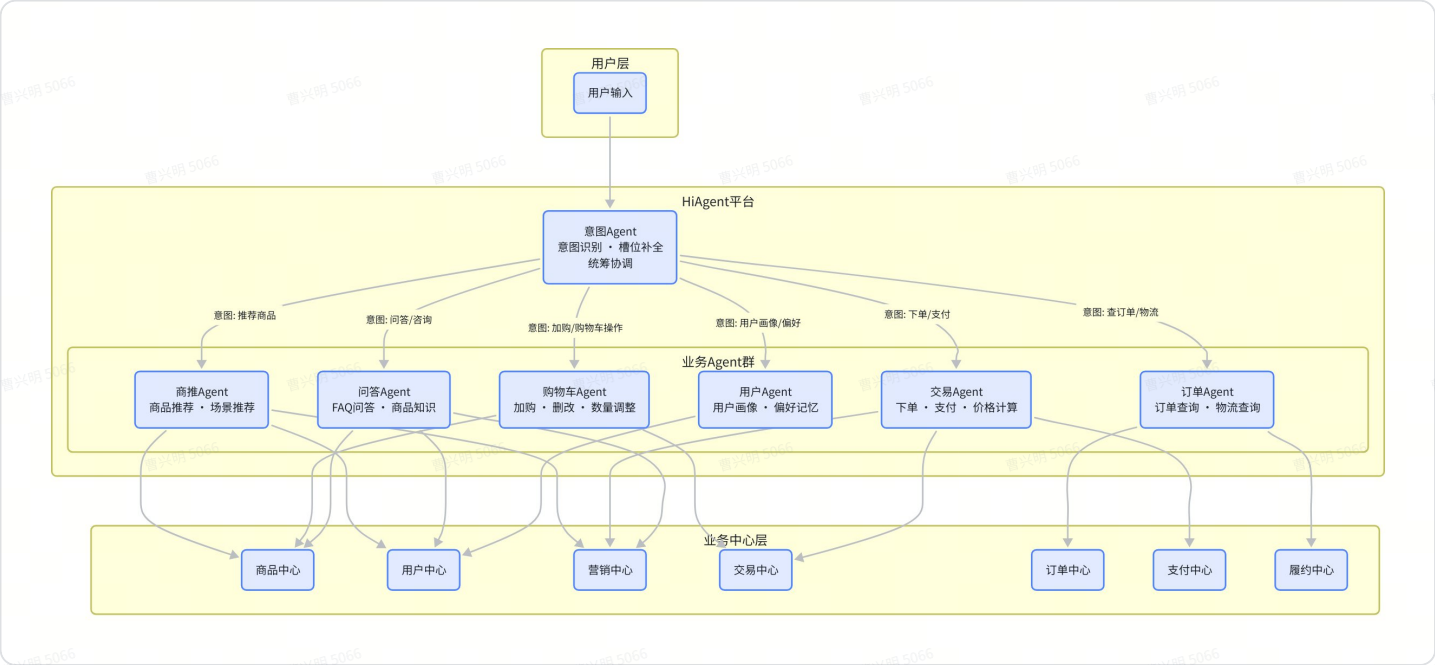
```
用户输入 → 意图Agent(识别意图)
    ↓
    ├─→ 商推Agent → 商品中心 → 返回推荐
    ├─→ 交易Agent → 购物车/订单 → 返回订单/物流
    ├─→ 问答Agent → 知识库 → 返回答案
    └─→ 用户Agent → 用户中心 → 返回画像
    └─→ 闲聊Agent → 豆包大模型 → 返回聊天内容

```

2.3 HiAgent 智能体协作关系图

协作模式：

模式	说明	示例场景
串行协作	前一个Agent完成后再调用下一个	选商品 → 加购 → 下单

并行协作	多个Agent同时处理不同任务	推荐商品 + 查询用户画像
嵌套协作	Agent内部再调用其他Agent	推荐商品 → 商品详情(问答Agent)



协作流程说明：

步骤	说明
1	用户发送自然语言请求（如"帮我推荐零食"）
2	意图Agent接收请求，进行意图识别和槽位补全
3	意图Agent根据识别结果，分发给对应的业务Agent
4	业务Agent调用相关业务中心完成具体功能
5	业务Agent返回结果给意图Agent
6	意图Agent整合结果，生成最终响应返回给用户

多Agent协作场景示例：

场景：用户说"帮我推荐零食并下单"

代码块

```
用户 ──────────────────────────────────────────────────> 意图Agent
                                                          |
                                                          ▼
                                                 识别为【商品推荐】+【创建订单】
```

```
                                        │
                                        ┌──────────────────────────┐
                                        ▼                          ▼
                                 商推Agent                  购物车Agent
                                        │                          │
                                        ▼                          ▼
                                 返回商品列表               返回加购成功
                                        │                          │
                                        └──────────────────────────┘
                                                   ▼
                                        交易Agent(创建订单)
                                                   │
                                                   ▼
                                        返回订单结果
                                                   │
                                                   ▼
用户 <──────────────────────────────────────────── 意图Agent(整合结果返回)
```

协作模式：

模式	说明	示例场景
串行协作	前一个Agent完成后再调用下一个	选商品 → 加购 → 下单
并行协作	多个Agent同时处理不同任务	推荐商品 + 查询用户画像
嵌套协作	Agent内部再调用其他Agent	推荐商品 → 商品详情(问答Agent)

header-row="true" column-widths="122,122,122,122,122,122">

步骤用户意图Agent商推Agent购物车Agent交易Agent

1帮我推荐零食并下单

2接收请求，识别为【商品推荐】+【创建订单】复合意图

3并行分发：推荐商品 + 加入购物车查询商品库加入购物车

4返回商品列表返回加购成功

5收到推荐+加购结果创建订单+支付

6返回订单结果

7整合推荐商品+订单详情，返回给用户

模式	说明	示例场景

串行协作	前一个Agent完成后再调用下一个	选商品 → 加购 → 下单
并行协作	多个Agent同时处理不同任务	推荐商品 + 查询用户画像
嵌套协作	Agent内部再调用其他Agent	推荐商品 → 商品详情(问答Agent)

3 详细交互流程

3.1 前端与后端 HTTP 交互

请求：

代码块

```
POST /api/ai-chat/send
Content-Type: application/json
Authorization: Bearer {token}

{
  "sessionId": "sess_xxx",
  "message": "帮我推荐零食",
  "messageType": "text",
  "userId": "user_123",
  "channel": "miniapp"
}

```

响应：

代码块

```
{
  "code": 0,
  "data": {
    "replyType": "card",
    "reply": {
      "type": "product_card_list",
      "content": { "cards": [...] }
    }
  }
}

```

3.2 网关与 HiAgent 交互

请求参数

参数	类型	必填	说明
LoginUserId	Long	否	会话ID登录用户ID
IntentName	String	是	意图名称
RequestTerminal	Object JSON	否	请求设备
RequestHeader	Object JSON	否	请求Header
RequestBody	Object JSON	否	请求Body

请求案例：

代码块

```
{
  LoginUserId: "213123131",
  // 1. 会话标识（从上一次响应中获取，如果是第一次则为空或新生成）
  AppConversationId: "sess_20260324_abc123",
  // 2. 用户指令（输入框内容 或 点击的建议气泡文案）
  Query: "帮我把购物车里所有商品数量都改成 2",
  //原始参数
  RequestTerminal: {
     hasClient: false,
     clientId: '1312313',
     client: {
         clientDeviceType: '',
         appVersion: '',
         province: '',
         deviceCode: '',
         platformId: ''
     }
  },
  // 3. 上下文环境（关键！让 AI 知道当前屏幕状态）
  context: {
    // 场景类型：告诉 AI 用户在哪个页面
    scene_type: "cart_preview", // 可选值: 'cart_preview', 'recommendation', 'order_detail'
    //当前意图
    //上一次
  }

}
```

响应参数：

`data.context` (上下文信息)

用于标识当前会话和场景，辅助前端做全局判断。

字段名	类型	必填	描述	示例值
appConversationId	String	是	当前会话的唯一标识，用于维持对话状态。	"sess_abc123"
scene_type	String	是	场景类型枚举，决定页面的基础形态。 可选值： - cart_preview: 购物车 - recommendation: 商品推荐	"cart_preview"
type	String	是	组件类型标识，前端据此选择渲染模板。	"card"
ai_text	String	是	AI 回复的自然语言文案，显示在头像旁的对话气泡中。	"好咧~商品已经全部加购，请查看。"
suggestions	Array[String]	否	快捷建议气泡列表。用户点击可直接发送对应指令。若为空则不显示。	["就按这个下单", "去购物车编辑"]

`data.businessData` (原始业务数据)

透传层。直接承载旧接口返回的所有页面参数。前端直接读取此对象中的字段进行数据渲染。

响应示例：

代码块

```
{
    "code": 0,
    "msg": "success",
    "data": {
        "context": {
            "appConversationId": "sess_a1b2c3d4e5f6",
            "sceneType": "cart_preview" ,
            "type": "text",
            "aiText": "好咧~商品已经全部加购，请查看。",
            "suggestions": ["就按这个下单", "去购物车编辑"],
            // 告诉前端当前是 'cart_preview' (图1) 还是 'recommendation' (图2)
        },
        "businessData": {// 旧接口返回的其他所有页面参数都放在这里
        }
    }
}
```

文字聊天 闲聊

文字聊天出参

```
{
    "code": 0,
    "msg": "success",
    "data": {
        "context": {
            "appConversationId": "sess_a1b2c3d4e5f6",
            "sceneType": "small_talk" ,
            "type": "text",
            "aiText": "哈哈你是想问哪里的天气呀～我这儿今天是大晴天，晒着太阳暖乎乎的超舒服😌你现在出门了嘛？要是天气好的话，逛完街逛累了可以带包来伊份的小零食去公园坐着吃，吹吹风嚼嚼脆货太惬意啦～"
        }
    }
}
```

思考消息 会替换消息 小伊正在分析你的需求 -> 小伊正在查找你的商品 -> 请稍等小伊正在推荐你的商品 最终只显示最后的

代码块

```
{
```

```
    "code": 0,
    "msg": "success",
    "data": {
        "context": {
            "appConversationId": "sess_a1b2c3d4e5f6",
            "sceneType": "cart_preview" ,
            "type": "thinking_step",
            "aiText": "小伊正在分析你的需求"
        }
    }
}
```

异常消息 哎呀抱歉了，出小差了

代码块

```
{
    "code": 0,
    "msg": "success",
    "data": {
        "context": {
            "appConversationId": "sess_a1b2c3d4e5f6",
            "sceneType": "cart_preview" ,
            "type": "error",
            "aiText": "小伊正在分析你的需求"
        }
    }
}
```

购物车

购物车出参

```
{
    "code": 0,
    "msg": "success",
    "data": {
        "context": {
            "appConversationId": "sess_a1b2c3d4e5f6",
            "sceneType": "cart_preview",
            "type": "card",
            "aiText": "好咧~商品已经全部加购，请查看。",
            "suggestions": [
                "就按这个下单",
                "去购物车编辑"
            ]
```

```
14          },
15          "businessData": {
16              "merchantList": [
17                  {
18                      "merchantId": 101,
19                      "merchantName": "来伊份",
20                      "merchantIconUrl": null,
21                      "merchantTelephone": null,
22                      "productList": [
23
24                      ],
25                      "summary": {
26                          "amount": 12.9,
27                          "beforeAmount": 0,
28                          "discount": 0,
29                          "vipSavedAmount": null,
30                          "vipDiscount": null,
31                          "totalNum": 1,
32                          "amountAll": 0,
33                          "checkAndUncheckNum": 0,
34                          "totalDeliveryFee": 0,
35                          "notIncludeFee": "不含运费"
36                      },
37                      "productGroups": [
38                          {
39                              "promotion": null,
40                              "summary": null,
41                              "productList": [
42                                  {
43                                      "mpId": 1158121307651895,
44                                      "productId": 1158121307651895,
45                                      "merchantId": 10000,
46                                      "name": "来伊份PUREC100%NFC葡萄汁1L",
47                                      "picUrl":
"https://static2.laiyifen.com/files/product-admin-
web/image/1d5a7648b03e440389bedf1ecf179361.jpg",
48                                      "num": 1,
49                                      "overLimitNum": 0,
50                                      "weight": 0,
51                                      "price": 12.9,
52                                      "promotionId": 0,
53                                      "tax": null,
54                                      "stockNum": 83,
55                                      "checked": 1,
56                                      "lackOfStock": 0,
57                                      "disabled": 0,
58                                      "disabledReason": null,
```

```
59                                  "promotionType": 0,
60                                  "promotionPrice": null,
61                                  "preferentialPrice": 0,
62                                  "promotionIconUrl": "",
63                                  "originalPrice": null,
64                                  "saleType": null,
65                                  "saleIconUrl": null,
66                                  "promotionUrl": null,
67                                  "tipsMsg": null,
68                                  "promotions": null,
69                                  "propertyTags": null,
70                                  "seriesParentId": null,
71                                  "itemType": 1,
72                                  "itemTxt": null,
73                                  "itemIconUrl": null,
74                                  "objectId": 0,
75                                  "tipsType": null,
76                                  "subtitle": "1瓶≈365颗夏黑葡萄",
77                                  "url60x60":
    "https://static2.laiyifen.com/files/product-admin-
    web/image/1d5a7648b03e440389bedf1ecf179361.jpg",
78                                  "url100x100":
    "https://static2.laiyifen.com/files/product-admin-
    web/image/1d5a7648b03e440389bedf1ecf179361.jpg",
79                                  "url160x160":
    "https://static2.laiyifen.com/files/product-admin-
    web/image/1d5a7648b03e440389bedf1ecf179361.jpg",
80                                  "url120x120":
    "https://static2.laiyifen.com/files/product-admin-
    web/image/1d5a7648b03e440389bedf1ecf179361.jpg",
81                                  "url220x220":
    "https://static2.laiyifen.com/files/product-admin-
    web/image/1d5a7648b03e440389bedf1ecf179361.jpg",
82                                  "url300x300":
    "https://static2.laiyifen.com/files/product-admin-
    web/image/1d5a7648b03e440389bedf1ecf179361.jpg",
83                                  "url400x400":
    "https://static2.laiyifen.com/files/product-admin-
    web/image/1d5a7648b03e440389bedf1ecf179361.jpg",
84                                  "url500x500":
    "https://static2.laiyifen.com/files/product-admin-
    web/image/1d5a7648b03e440389bedf1ecf179361.jpg",
85                                  "url800x800":
    "https://static2.laiyifen.com/files/product-admin-
    web/image/1d5a7648b03e440389bedf1ecf179361.jpg",
86                                  "paidProductIcon": null,
87                                  "isCollected": 0,
```

```
                                "isDirectSupply": 1,
                                "gift": false
                            }
                        ],
                        "giftProductList": [

                        ]
                    }
                ],
                "sendOutMoneyUpFlag": null,
                "sendOutMoneyUp": null,
                "freeShipping": null,
                "overseas": null,
                "channelCode": "IF01_115"
            }
        ],
        "summary": {
            "amount": 12.9,
            "beforeAmount": 12.9,
            "discount": 0,
            "vipSavedAmount": null,
            "vipDiscount": null,
            "totalNum": 1,
            "amountAll": 0,
            "checkAndUncheckNum": 0,
            "totalDeliveryFee": 0,
            "notIncludeFee": "不含运费"
        },
        "failureProducts": [

        ],
        "showPaidMember": true,
        "showPaidMemberType": 0,
        "paidMemberIcon": "https://static4.laiyifen.com/files/sds-admin-web/other/data/upload/a6fcbe131acc48cd99702891af8fca9f.png",
        "paidMemberDesc": "0.64",
        "ptPaidMemberDesc": null,
        "shippingRules": [
            {
                "lines": [
                    "实付满68 元包邮（港澳台及特殊地区除外），未满按首重 1kg 计费：",
                    "江浙沪：首重 5 元，续重 1 元 /kg",
                    "华北 / 华东 / 粤闽 / 湖北：首重 8 元，续重 5 元 /kg",
                    "华中 / 川渝 / 晋陕豫 / 桂琼：首重 10 元，续重 7 元 /kg",
                    "其他地区请见运费详情"
                ]
```

```
                }
            ],
            "freeShippingRule": "满68元包邮",
            "shippingCmsPath": "https://m.test.laiyifen.com/cms/view/h5/article/1678999047219200.html",
            "paidMemberJumpUrl": "https://m.laiyifen.com/hj/index?tracecode=f72675f3a39e467aaf6ea023b52353a1"
        }
    }
}
```

确认订单页

确认订单页出参

```
{
    "code": 0,
    "msg": "success",
    "data": {
        "context": {
            "appConversationId": "sess_a1b2c3d4e5f6",
            "sceneType": "order_init" ,
            "type": "card",
            "aiText": "请确认订单信息，如有修改告诉我",
            "suggestions": []
        },
        "businessData": {
                "orderCode": null,
                "receiver": {
                        "receiverId": 5219656,
                        "name": "王涛",
                        "mobile": "17743211234",
                        "provinceId": null,
                        "provinceName": "上海",
                        "provinceCode": "310000",
                        "cityId": null,
                        "cityName": "上海市",
                        "cityCode": "310100",
                        "areaId": null,
                        "areaName": "松江区",
                        "areaCode": "310117",
                        "detailAddress": "九亭地铁站",
                        "postCode": null,
                        "exactAddress": null,
                        "sex": null,
                        "realName": "王涛",
```

```
32                          "identityCardNumber": null,
33                          "isDefault": 1,
34                          "longitude": 121.326325,
35                          "latitude": 31.13026
36                  },
37                  "payments": [{
38                          "paymentId": 1,
39                          "name": "网上支付",
40                          "selected": 1,
41                          "isAvailable": 1,
42                          "unavailableReason": null,
43                          "availableCouponNum": 0,
44                          "payIconUrl": null
45                  }],
46                  "merchantList": [{
47                          "id": 0,
48                          "merchantId": 101,
49                          "merchantName": "来伊份",
50                          "merchantIconUrl": null,
51                          "productList": [{
52                                  "mpId": 1158121242601408,
53                                  "name": "蟹香豆瓣78g",
54                                  "picUrl":
    "https://static2.laiyifen.com/files/product-admin-
    web/image/389287d917704ecf9155f6dccbce5ccd.jpg",
55                                  "num": 1,
56                                  "weight": 0.0,
57                                  "price": 5.90,
58                                  "tax": null,
59                                  "stock": null,
60                                  "mpType": 0,
61                                  "productAmount": 5.90,
62                                  "grossWeight": null,
63                                  "netWeight": null,
64                                  "freightAttribute": null,
65                                  "merchantProdVolume": null,
66                                  "isGift": 2,
67                                  "exciseTaxAmount": null,
68                                  "incrementTaxAmount": null,
69                                  "customsDutiesAmount": null,
70                                  "standard": null,
71                                  "propertyTags": [],
72                                  "promotionType": null,
73                                  "isVatInvoice": null,
74                                  "purchaseType": null,
75                                  "purchaseIcon": null,
76                                  "typeOfProduct": null,
```

```
                                "usedYidou": null,
                                "yidouAccount": null,
                                "supportInvoiceType": 0,
                                "isSupportEInvoice": 0,
                                "originalPrice": null,
                                "url60x60": "https://static2.laiyifen.com/files/product-admin-web/image/389287d917704ecf9155f6dccbce5ccd.jpg",
                                "url100x100": "https://static2.laiyifen.com/files/product-admin-web/image/389287d917704ecf9155f6dccbce5ccd.jpg",
                                "url160x160": "https://static2.laiyifen.com/files/product-admin-web/image/389287d917704ecf9155f6dccbce5ccd.jpg",
                                "url120x120": "https://static2.laiyifen.com/files/product-admin-web/image/389287d917704ecf9155f6dccbce5ccd.jpg",
                                "url220x220": "https://static2.laiyifen.com/files/product-admin-web/image/389287d917704ecf9155f6dccbce5ccd.jpg",
                                "url300x300": "https://static2.laiyifen.com/files/product-admin-web/image/389287d917704ecf9155f6dccbce5ccd.jpg",
                                "url400x400": "https://static2.laiyifen.com/files/product-admin-web/image/389287d917704ecf9155f6dccbce5ccd.jpg",
                                "url500x500": "https://static2.laiyifen.com/files/product-admin-web/image/389287d917704ecf9155f6dccbce5ccd.jpg",
                                "url800x800": "https://static2.laiyifen.com/files/product-admin-web/image/389287d917704ecf9155f6dccbce5ccd.jpg"
                            }],
                            "totalNum": 1,
                            "totalWeight": null,
                            "amount": 5.90,
                            "tax": null,
                            "deliveryFee": 5.00,
                            "overseaModeName": null,
                            "overseaModeId": null,
                            "remark": null,
                            "usedYidou": null,
                            "yidouAccount": null,
                            "channelCode": "IF01_115",
                            "isToday": null,
                            "time": null
                    }],
```

```
106                 "merchantDeliveryModeList": [{
107                         "id": "0",
108                         "merchantId": 101,
109                         "merchantName": "来伊份",
110                         "deliveryModeList": [{
111                                 "isDefault": 1,
112                                 "code": null,
113                                 "name": "商城自营配送",
114                                 "freight": null,
115                                 "isTakeTheir": null,
116                                 "pickSiteId": null,
117                                 "pickMobile": null,
118                                 "pickName": null,
119                                 "pickAddress": null,
120                                 "promiseDate": null
121                         }],
122                         "deliveryFee": 5.00
123                 }],
124                 "totalNum": 1,
125                 "amount": 10.90,
126                 "totalWeight": 0,
127                 "totalDeliveryFee": 5.00,
128                 "deliveryFeeDiscount": 0,
129                 "deliveryFeeNeedPay": 5.00,
130                 "deliveryFeeTotal": 5.00,
131                 "promotionSavedAmount": 0,
132                 "productAmount": 5.90,
133                 "giftCardAmount": 0,
134                 "couponAmount": 0,
135                 "shippingFeeCouponAmount": null,
136                 "totalTax": 0,
137                 "vipSavedAmount": 0,
138                 "productTaxes": [],
139                 "userCertification": {
140                         "needCertification": false,
141                         "type": null,
142                         "name": null,
143                         "certificateNo": null,
144                         "account": null,
145                         "temp": null
146                 },
147                 "oversea": false,
148                 "overseaPurchase": 0,
149                 "isIdCardVerifyRequired": 0,
150                 "orderInvoice": {
151                         "merchantSupportInvoiceType": 3,
152                         "merchantSupportInvoiceMessage": "",
```

```
153                 "invoice": {
154                         "invoiceType": 0,
155                         "invoiceTitleType": null,
156                         "invoiceTitleContent": null,
157                         "isNeedDetails": null,
158                         "invoiceContentId": null,
159                         "invoiceContent": null,
160                         "unitName": null,
161                         "taxpayerIdentificationCode": null,
162                         "registerAddress": null,
163                         "registerPhone": null,
164                         "bankDeposit": null,
165                         "bankAccount": null,
166                         "receiverName": null,
167                         "receiverMobile": null,
168                         "receiverCountry": null,
169                         "receiverProvince": null,
170                         "receiverCity": null,
171                         "receiverArea": null,
172                         "receiverAddress": null,
173                         "auditStatus": null,
174                         "takerMobile": null,
175                         "takerEmail": null,
176                         "invoiceMode": null
177                 },
178                 "invoiceContentList": [{
179                         "invoiceContentId": 1,
180                         "invoiceContentValue": "食品"
181                 }],
182                 "soInviceConfig": [{
183                         "invoiceMode": 1,
184                         "invoiceModeName": "电子发票"
185                 }, {
186                         "invoiceMode": 2,
187                         "invoiceModeName": "纸质发票"
188                 }],
189                 "isSupportEInvoice": 0
190         },
191         "remark": null,
192         "deliveryTime": null,
193         "giftCard": null,
194         "coupons": null,
195         "allCoupon": {
196                 "orderCoupons": [],
197                 "selected": 0,
198                 "availableQuantity": 0,
199                 "selectedCount": 0,
```

```
                        "preferentialPrice": 0,
                        "shippingFeeOrderCoupons": [],
                        "shippingFeeSelected": 1,
                        "availableShippingFeeQuantity": null,
                        "selectedShippingFeeCount": 0,
                        "preferentialShippingFeePrice": null
                },
                "businessType": 0,
                "points": null,
                "brokerage": null,
                "expenseList": [{
                        "label": "商品总价",
                        "value": "5.90",
                        "operator": "",
                        "iconUrl": null
                }, {
                        "label": "运费合计",
                        "value": "5.00",
                        "operator": "+",
                        "iconUrl": null
                }],
                "presell": null,
                "countdown": 0,
                "countdownTheme": null,
                "orderUCard": null,
                "orderECard": null,
                "error": null,
                "receiverStatus": 0,
                "usedYidou": null,
                "yidouAccount": null,
                "canUseCoupons": true,
                "canUsePoints": true,
                "canUseBrokerage": true,
                "canUseUCard": true,
                "canUseECard": true,
                "canUsePostageCoupons": true,
                "postageCouponAmount": 0,
                "allPostageCoupon": null,
                "showPaidMember": true,
                "paidMemberIcon": "https://static1.laiyifen.com/files/paid-member/image/crown/coupons_icon_Black_stamps_2x.png",
                "paidMemberDesc": "0.29",
                "showGlobalNotice": false
        }
}
}
```

订单列表

订单列表出参


```
{
    "code": 0,
    "msg": "success",
    "data": {
        "context": {
            "appConversationId": "sess_a1b2c3d4e5f6",
            "sceneType": "order_list" ,
            "type": "card",
            "aiText": "请确认订单信息，如有修改告诉我",
            "suggestions": []
        },
        "businessData":  {
                "orderList": [{
                        "orderCode": "272202603121636219938073 3",
                        "orderStatus": 0,
                        "orderStatusName": "已取消",
                        "subItemTxt": null,
                        "paymentAmount": 27.30,
                        "totalCount": 1,
                        "merchantLogo": "https://static1.laiyifen.com/test-files/order-app-admin-web/image/appLogo/d6fe69ecc2104f2a924e46e52cc7a188.png",
                        "merchantName": "西域大巴扎特产店",
                        "merchantRouting": null,
                        "merchantPath": null,
                        "merchantId": null,
                        "tradeType": 1,
                        "groupTag": "",
                        "productList": [{
                                "productCname": "西域皇后酱牛肉100g*1袋",
                                "productPicPath": "https://static4.laiyifen.com/files/product-admin-web/image/3f92235141c244e5aff3e44070ce4632.jpg",
                                "num": 1.000,
                                "weight": null,
                                "goodReceiverMobile": null,
                                "orderExtra": null,
                                "numWeightDes": "x1"
                        }],
                        "operationButtonList": [{
                                "operationCode": "deleteOrder",
                                "operationName": "删除订单",
                                "url": null,
                                "path": null,
```

```
                                "routing": "deleteOrder",
                                "isHighlight": 0
                        }, {
                                "operationCode": "bugAgain",
                                "operationName": "再来一单",
                                "url": null,
                                "path": null,
                                "routing": "bugAgain",
                                "isHighlight": 0
                        }],
                        "trackInfo": {
                                "eventName": "订单已取消",
                                "eventTime": "2026-03-12T09:37:05.293+00:00",
                                "createTimeStr": "2026-03-12 17:37:05",
                                "trackRouteUrl": "https://m.laiyifen.com/order/logistics/index?orderCode=272202603121636219938073 3",
                                "relationNo": "272202603121636219938073 3"
                        },
                        "isMultiPackage": false,
                        "packageCode": null,
                        "channel": "115",
                        "orderDetailRouting": "https://m.laiyifen.com/order/orderdetail?orderCode=272202603121636219938073 3",
                        "addressId": null,
                        "orderDetailPath": "https://m.laiyifen.com/order/orderdetail?orderCode=272202603121636219938073 3",
                        "subscript": "",
                        "expectDeliveryTime": null,
                        "delivery": null,
                        "popOrder": true,
                        "updateAvailable": 2,
                        "updateAvailableMsg": "不可修改",
                        "shopCode": "71945L"
                }],
                "totalCount": 324
        }
    }
}
```


订单详情

代码块


```
{
    "code": 0,
```

```
    "msg": "success",
    "data": {
        "context": {
            "appConversationId": "sess_a1b2c3d4e5f6",
            "sceneType": "order_detail" ,
            "type": "card",
            "aiText": "请确认订单信息，如有修改告诉我",
            "suggestions": []
        },
        "businessData":  {
                "id": null,
                "orderNo": null,
                "totalAmount": null,
                "payAmount": null,
                "payType": null,
                "discountAmount": null,
                "deliveryTypeCode": "D01",
                "createTime": "2026-01-15 22:49:48",
                "statusDesc": "已取消",
                "descInfo": "订单超时未支付系统取消",
                "expectPickupTime": null,
                "totalQuantity": 1,
                "tradeStatus": 0,
                "paymentStatus": null,
                "cancelReasonDesc": "订单超时未支付系统取消",
                "cancelType": 2,
                "orderAddress": {
                        "extData": null,
                        "goodReceiverAddress": "九亭地铁站",
                        "goodReceiverArea": "松江区",
                        "goodReceiverAreaId": "310117",
                        "goodReceiverCity": "上海市",
                        "goodReceiverCityId": "310100",
                        "goodReceiverMobile": "17743211234",
                        "goodReceiverName": "王涛",
                        "goodReceiverPostcode": null,
                        "goodReceiverProvince": "上海",
                        "goodReceiverProvinceId": "310000",
                        "goodReceiverAddressDetail": "上海上海市松江区",
                        "shopCode": null,
                        "shopName": null
                },
                "storeList": null,
                "goodsList": [{
                        "skuName": "53°醉爱6#酱香酒500ml 升级版2025",
                        "pic": "https://static5.laiyifen.com/files/product-admin-web/image/80adb90821024896975fbcd711d80f9d.jpg",
```

```
                        "skuQuantity": 1,
                        "skuCode": "31339",
                        "unitCode": "BOT",
                        "skuPrice": null,
                        "discountPrice": 799.0,
                        "saleUnitName": "瓶",
                        "smallUnitName": "瓶",
                        "smallPrice": null,
                        "smallDiscountPrice": null,
                        "className": "酒饮系列",
                        "actualPrice": 799.0
                }],
                "amountDescList": [],
                "recordCode": "89260115938001",
                "tmsRecordCode": null,
                "commissionAmount": null,
                "superiorCommissionAmount": null,
                "invoiceId": null,
                "orderType": null,
                "invoiceDTO": {
                        "invoiceStatus": 4,
                        "invoiceText": "当前状态不支持开票",
                        "url": null
                },
                "cancelTime": null,
                "labDTOs": []
        }
    }
}
```


物流轨迹

代码块


```
{
    "code": 0,
    "msg": "success",
    "data": {
        "context": {
            "appConversationId": "sess_a1b2c3d4e5f6",
            "sceneType": "track_detail",
            "type": "card",
            "aiText": "请确认订单信息，如有修改告诉我",
            "suggestions": []
        },
        "businessData":  {
```

```
                "receiverImageUrl": null,
                "receiverAddress": "九亭地铁站",
                "receiverPhone": "177****1234",
                "currentTrackDTO": {
                        "currentImage": "https://static1.laiyifen.com/files/order-app-admin-web/image/appLogo/364c068dab3a43c8af93a190d8f12542.png",
                        "currentEventDesc": "伊仔全力为您服务中~",
                        "officialTelephone": "95311",
                        "packageCode": null
                },
                "trackPointList": [{
                        "pointName": null,
                        "pointImageUrl": null,
                        "trackEventList": [{
                                "eventDesc": "订单已提交",
                                "eventTime": "2026-03-26 16:52:59"
                        }]
                }]
        }
    }
}
```


3.3 HiAgent 与业务中心交互

代码块

```
HiAgent → 商品中心: /api/goods/search
HiAgent → 用户中心: /api/user/profile
HiAgent → 交易中心: /api/cart/add
HiAgent → 订单中心: /api/order/create
HiAgent → 支付中心: /api/payment/create

```

---

4 接口详细设计

4.1 对话消息发送接口

接口信息：

项目	内容
路径	`/api/ai-chat/send`
方法	POST
鉴权	是

请求参数：

参数	类型	必填	说明
sessionId	String	是	会话ID
message	String	是	消息内容
messageType	String	是	text/image/voice
userId	String	是	用户ID
channel	String	是	miniapp/h5/ios/android
clientInfo	Object	否	客户端信息
context	Object	否	上下文信息

请求示例：

代码块

```
{
  "sessionId": "sess_a1b2c3d4e5f6",
  "message": "帮我推荐适合送长辈的零食礼盒",
  "messageType": "text",
  "userId": "user_123456",
  "channel": "miniapp",
  "clientInfo": {
    "platform": "wechat",
    "deviceId": "device_xxxxx",
    "appVersion": "2.1.0"
  },
```

```
    "context": {
      "currentPage": "chat",
      "lastIntent": "商品推荐"
    }
  }

```

响应参数：

参数	类型	说明
code	Integer	0=成功
data.sessionId	String	会话ID
data.replyType	String	回复类型
data.reply	Object	回复内容

响应示例：

代码块

```
{
  "code": 0,
  "message": "success",
  "data": {
    "sessionId": "sess_a1b2c3d4e5f6",
    "messageId": "msg_xxx",
    "replyType": "card",
    "reply": {
      "type": "product_card_list",
      "content": {
        "cards": [
          {
            "id": "card_001",
            "type": "product",
            "product": {
              "id": "prod_1001",
              "name": "来伊份坚果礼盒1588g",
              "image": "https://cdn.laiyifen.com/images/1001.jpg",
              "price": 128.00,
              "originalPrice": 168.00,
              "tags": ["送礼佳选", "限时优惠"],
              "stock": 100
            },
```

```
            "actions": [
              { "type": "add_cart", "label": "加购", "value": "prod_1001" },
              { "type": "buy_now", "label": "直接买", "value": "prod_1001" }
            ]
          }
        ]
      },
      "text": "为您推荐以下几款适合送长辈的零食礼盒：",
      "suggestions": ["换一个", "看看优惠活动", "我的购物车"]
    },
    "slots": { "scene": "送礼", "target": "长辈" },
    "intent": "商品推荐",
    "confidence": 0.95
  }
}

```

4.2 会话历史查询接口

接口信息：

项目	内容
路径	`/api/ai-chat/history`
方法	GET
鉴权	是

请求参数：

参数	类型	必填	说明
sessionId	String	是	会话ID
page	Integer	否	页码，默认1
pageSize	Integer	否	每页条数，默认20

响应示例：

```
代码块
```

```json
{
  "code": 0,
  "data": {
    "sessionId": "sess_xxx",
    "sessionInfo": {
      "createdAt": 1709999000000,
      "updatedAt": 1709999999999,
      "lastIntent": "商品推荐",
      "status": "active"
    },
    "messages": [
      {
        "messageId": "msg_001",
        "role": "user",
        "content": "帮我推荐零食",
        "timestamp": 1709999900000,
        "intent": "商品推荐",
        "slots": { "category": "零食" }
      },
      {
        "messageId": "msg_002",
        "role": "assistant",
        "content": "好的，请问您有什么口味偏好？",
        "timestamp": 1709999910000,
        "replyType": "text"
      }
    ]
  }
}

```

4.3 用户画像查询接口

接口信息：

项目	内容
路径	`/api/ai-user/profile`
方法	GET
鉴权	是

请求参数：

参数	类型	必填	说明
userId	String	是	用户ID
profileTypes	String	否	preference,allergy,behavior

响应示例：

代码块

```
{
  "code": 0,
  "data": {
    "userId": "user_123456",
    "profiles": {
      "preference": {
        "口味偏好": ["低糖", "少盐"],
        "预算范围": "50-100元",
        "品牌偏好": ["来伊份"]
      },
      "allergy": {
        "忌口": ["坚果", "海鲜"],
        "过敏原": ["花生"]
      },
      "behavior": {
        "常购品类": ["零食", "坚果"],
        "购买频次": "每周1-2次",
        "平均客单价": "80元"
      }
    }
  }
}

```

4.4 购物车操作接口

接口信息：

项目	内容

路径	`/api/cart/add`
方法	POST
鉴权	是

请求参数：

参数	类型	必填	说明
sessionId	String	是	会话ID
userId	String	是	用户ID
productId	String	是	商品ID
skuId	String	否	SKU ID
quantity	Integer	是	数量
source	String	是	ai_chat/normal

请求示例：

代码块

```
{
  "sessionId": "sess_a1b2c3d4e5f6",
  "userId": "user_123456",
  "productId": "prod_1001",
  "skuId": "sku_1001_1",
  "quantity": 1,
  "source": "ai_chat"
}

```

响应示例：

代码块

```
{
  "code": 0,
  "data": {
    "cartId": "cart_xxx",
    "cartItems": [
      {
```

```
7          "itemId": "item_001",
8          "productId": "prod_1001",
9          "productName": "来伊份坚果礼盒",
10         "skuInfo": "经典款",
11         "price": 128.00,
12         "quantity": 1,
13         "subtotal": 128.00
14       }
15     ],
16     "cartSummary": {
17       "totalItems": 1,
18       "totalQuantity": 1,
19       "totalAmount": 128.00,
20       "payableAmount": 128.00
21     },
22     "suggestions": ["再买38元免运费", "热门商品推荐"]
23   }
24 }
25
```

4.5 订单创建接口

接口信息：

项目	内容
路径	`/api/order/create`
方法	POST
鉴权	是

请求参数：

参数	类型	必填	说明
sessionId	String	是	会话ID
userId	String	是	用户ID
items	Array	是	商品列表
deliveryInfo	Object	是	配送信息

deliveryInfo.type	String	是	self_pickup/delivery
deliveryInfo.storeId	String	否	门店ID
deliveryInfo.addressId	String	否	地址ID
couponId	String	否	优惠券ID
paymentMethod	String	是	wechat_pay/alipay

请求示例：

代码块

```
{
  "sessionId": "sess_a1b2c3d4e5f6",
  "userId": "user_123456",
  "items": [
    { "productId": "prod_1001", "skuId": "sku_1001_1", "quantity": 1, "price": 128.00 }
  ],
  "deliveryInfo": {
    "type": "self_pickup",
    "storeId": "store_001"
  },
  "paymentMethod": "wechat_pay"
}

```

响应示例：

代码块

```
{
  "code": 0,
  "data": {
    "orderId": "order_xxx",
    "orderNo": "LYF20260311200001",
    "orderStatus": "pending_payment",
    "orderSummary": {
      "goodsAmount": 128.00,
      "discountAmount": 10.00,
      "freight": 0,
      "totalAmount": 118.00
    },
    "paymentInfo": {
      "paymentMethod": "wechat_pay",
```

```
        "payableAmount": 118.00,
        "expireTime": 1800000
      },
      "confirmationCard": {
        "type": "order_confirm",
        "title": "订单确认",
        "totalAmount": 118.00,
        "actions": [
          { "type": "confirm", "label": "确认支付" },
          { "type": "cancel", "label": "取消" }
        ]
      }
    }
  }

```

4.6 订单查询接口

接口信息：

项目	内容
路径	`/api/order/query`
方法	POST
鉴权	是

请求参数：

参数	类型	必填	说明
sessionId	String	是	会话ID
message	String	是	用户消息
userId	String	是	用户ID
context.queryType	String	否	list/status/logistics
context.orderNo	String	否	订单号

响应示例：

代码块

```
{
  "code": 0,
  "data": {
    "replyType": "card",
    "reply": {
      "type": "order_status",
      "content": {
        "orderNo": "LYF20260311200001",
        "orderStatus": "delivering",
        "statusText": "配送中",
        "logistics": {
          "courier": "顺丰速运",
          "trackingNo": "SF1234567890",
          "estimatedArrival": "今天 18:00 前"
        }
      },
      "suggestions": ["查看订单详情", "联系客服"]
    }
  }
}

```

4.7 搜索商品接口

接口信息：

项目	内容
路径	`/api/goods/search`
方法	POST

请求参数：

参数	类型	必填	说明
sessionId	String	是	会话ID
message	String	是	搜索内容
userId	String	是	用户ID

channel	String	是	渠道

响应示例：

代码块

```
{
  "code": 0,
  "data": {
    "replyType": "card",
    "reply": {
      "type": "product_card_list",
      "content": { "cards": [...] }
    }
  }
}

```

4.8 问答FAQ接口

接口信息：

项目	内容
路径	`/api/ai-chat/send` (问答场景)
方法	POST

请求参数：

参数	类型	必填	说明
sessionId	String	是	会话ID
message	String	是	问题内容
userId	String	是	用户ID

响应示例：

代码块

```
{
  "code": 0,
  "data": {
    "replyType": "mixed",
    "reply": {
      "type": "faq_answer",
      "content": {
        "question": "会员有什么权益？",
        "answer": "来伊份会员权益：\n\n🎟️ 积分权益：消费1元积1分\n💰 折扣权益：银卡9.5折/金卡9折/黑金8.8折\n🎁 专属活动：会员日/生日月双倍积分"
      },
      "suggestions": ["如何成为会员", "积分怎么使用"]
    }
  }
}

```

4.9 闲聊接口

接口信息：

项目	内容
路径	`/api/ai-chat/send` (闲聊场景)
方法	POST

响应示例：

代码块

```
{
  "code": 0,
  "data": {
    "replyType": "mixed",
    "reply": {
      "type": "casual_chat",
      "content": {
        "text": "您好！很高兴为您服务～有什么可以帮您的吗？",
        "emotion": "friendly"
      },
      "suggestions": ["帮我推荐零食", "有什么优惠活动"]
    }
  }
```

```
}

```

4.10 偏好写入接口

请求参数：

参数	类型	必填	说明
sessionId	String	是	会话ID
message	String	是	偏好内容
userId	String	是	用户ID
context.preferenceType	String	否	allergy/taste/budget
context.preferenceValue	String	否	偏好值

响应示例：

代码块

```
{
  "code": 0,
  "data": {
    "replyType": "card",
    "reply": {
      "type": "preference_confirm",
      "content": {
        "title": "记录忌口偏好",
        "summary": "花生",
        "actions": [
          { "type": "confirm_long", "label": "长期有效" },
          { "type": "confirm_session", "label": "仅本次" }
        ]
      }
    }
  }
}

```

---

5 后台服务间交互

5.1 网关与 HiAgent 交互

请求：

代码块

```
POST /hiagent/api/v1/chat/send
Header:
  X-Request-ID: req_xxx
  X-User-ID: user_123
  X-Channel: miniapp

```

5.2 HiAgent 与业务中心交互

业务中心	接口	注意事项
商品中心	/api/goods/search	处理上下架状态
用户中心	/api/user/profile	敏感信息脱敏
营销中心	/api/promotion/coupons	处理活动时效
交易中心	/api/cart/add, /api/order/create	保证事务一致性
订单中心	/api/order/query	处理状态机流转
支付中心	/api/payment/create	处理回调通知

---

6 亮点设计

6.1 意图识别设计

多标签意图分类：

分类	意图标签	说明
交易	选品推荐、加购、下单支付、用券	购物相关
售后	查订单、物流	订单售后

咨询	FAQ、商品知识	问答咨询
用户	偏好写入	用户画像
闲聊	闲聊/兜底	寒暄聊天

槽位补全机制：

1. 从消息中提取关键实体
2. 映射到标准槽位标签
3. 优先使用历史默认值自动补全
4. 无法补全则触发追问

6.2 历史会话设计

会话存储结构：

代码块

```
{
  "sessionId": "sess_xxx",
  "userId": "user_123",
  "context": {
    "currentIntent": "商品推荐",
    "currentSlots": { "场景": "送礼", "对象": "长辈" },
    "selectedProducts": [...],
    "cartInfo": { "itemCount": 2, "totalAmount": 200 }
  },
  "messages": [...]
}

```

上下文继承： 新会话可继承历史上下文，避免重复询问

6.3 接口对接设计

统一 API 网关功能：

- 路由转发
- 鉴权认证 (JWT)
- 限流熔断
- 日志监控
- 协议转换

- 缓存加速
- 安全防护

HiAgent 接入规范：

- 协议: HTTPS
- 认证: AppKey + AppSecret 签名 / JWT Token
- 超时: 30s (可配置)
- 重试: 失败重试 3 次，指数退避

前端和网关交互

1. 对话购物车出参

代码块

```
{
  "code": 0,
  "msg": "success",
  "data": {
    "pageType": "cart_preview_and_checkout",
    "sessionId": "sess_a1b2c3d4e5f6",
    "messageId": "msg_cart_summary_001",
    "replyType": "card",
    "reply": {
      "type": "cart_summary_card",
      "content": {
        "title": "已加购商品: 8件",
        "subtitle": "去购物车 > ",
        "items": [
          {
            "mpId": "mp_2001",
            "productName": "伊份高蛋白脆腿肉肠 (孜然烧烤味) 35g",
            "subTitle": "孜然烤肠新口味 | 二次烘烤锁鲜",
            "image": "https://cdn.laiyifen.com/images/2001.jpg",
            "price": 8.80,
            "quantity": 1,
            "subtotal": 8.80,
            "tags": ["热销爆款"]
          },
          {
```

```
26            "mpId": "mp_2002",
27            "productName": "伊份梅好 无核陈皮梅45g",
28            "subTitle": "果肉细腻，酸中带甜",
29            "image": "https://cdn.laiyifen.com/images/2002.jpg",
30            "price": 8.80,
31            "quantity": 1,
32            "subtotal": 8.80,
33            "tags": ["经典回味"]
34          },
35          {
36            "mpId": "mp_2003",
37            "productName": "伊份梅好 无核陈皮梅45g",
38            "subTitle": "果肉细腻，酸中带甜",
39            "image": "https://cdn.laiyifen.com/images/2002.jpg",
40            "price": 8.80,
41            "quantity": 1,
42            "subtotal": 8.80,
43            "tags": ["经典回味"]
44          },
45          {
46            "mpId": "mp_2004",
47            "productName": "伊份梅好 无核陈皮梅45g",
48            "subTitle": "果肉细腻，酸中带甜",
49            "image": "https://cdn.laiyifen.com/images/2002.jpg",
50            "price": 8.80,
51            "quantity": 1,
52            "subtotal": 8.80,
53            "tags": ["经典回味"]
54          }
55        ],
56        "summary": {
57          "totalItems": 8,
58          "totalAmount": 182.40,
59          "discountTip": "满¥68包邮 ✓",
60          "freeShippingThreshold": 68.00,
61          "isFreeShipping": true
62        }
63      },
64      "text": "好咧～商品已经全部加购，请查看。\n您也可以直接跟我说「就按这个下单」，我马上帮您搞定",
65      "suggestions": [
66        "就按这个下单",
67        "去购物车编辑",
68        "再加点别的"
69      ]
70    }
71  }
```

```
}
```

7 附录

7.1 数据字典

错误编码：

序 号	编 码	错误编码说明
001	2000400	交易智能体全局参数校验失败，参数为空和格式不正确

会话状态：

状态码	名称	说明
0	active	活跃会话
1	waiting	等待输入
2	processing	处理中
3	closed	已关闭

消息类型：

类型	说明
text	文本消息
image	图片消息
voice	语音消息
button_event	按钮事件

回复类型：

类型	说明
text	文字回复
card	卡片回复
action	动作回复
mixed	混合回复

7.2 版本记录

版本	日期	修改人	内容
V1.0	2026-03-11	JaNa	初始版本

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